

Aditya Kumar Panigrahy

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Education

- Vellore Institute of Technology** Vellore, Tamil Nadu
Computer Science and Engineering with Specialization in Internet of Things; CGPA: 8.48 Sept 2022 - May 2026
- D.A.V. Public School** Brahmapur, Odisha
Higher Education in Computer Science; Percentage: 87.4% March 2019 - May 2021

Skills Summary

- Languages: Java, JavaScript, OOP, SQL, Python, TypeScript
- Frameworks: Node.js, React.js, Next.js, Express.js, Tailwind CSS, shadcn/ui
- Tools / Databases: Git, Neon, MySQL, SQLite, MongoDB, SQL Server, Redis, MongoDB Atlas, Pinecone
- Platforms: Web, Windows, Arduino, Raspberry Pi, AWS, Modal, Firebase, Supabase, Clerk, Vercel, Kuberns
- AI Engineering & LLM Architecture: Gemini 1.5 Pro/Flash, GPT-4o, Qwen2.5-7B-Instruct, stabilityai/sdxl-turbo, Langchain

Experience

- Kuberns** Remote
UI/UX Intern Nov 2024 - April 2025
 - Redesigned IA flows and multi-step onboarding using usability testing, pain-point mapping, and iterative feedback loops, introducing interactive UX improvements. This reduced task time by 25% and drove +24% organic growth, higher retention, and stronger product adoption.
- NAPX** Remote
UI/UX Intern June 2024 - July 2024
 - Drove organic growth through SEO-focused content, performance optimization, and conversion-led UX improvements, while building a high-conversion e-commerce platform ui using React + Tailwind. This resulted in +40% traffic growth, faster client acquisition, and improved checkout conversions.

Projects

- PDF Chat** – *Next.js, Tailwind CSS, shadcn/ui, MongoDB, LangChain, OpenAI, Pinecone, Clerk*
 - A full-stack PDF assistant using anthropic level chunking, hybrid retrieval (dense + sparse embeddings), and threshold-based reranking to enable accurate conversational search over large documents.
 - Optimized retrieval pipeline with LangChain and Pinecone by tuning chunk granularity and implementing rerank filtering, reducing query latency by 50% while improving contextual relevance of responses.
- Swara** – *Python, Next.js, Neon, Modal, AWS, Polar, Inngest, TypeScript, ACE-Step, Qwen*
 - A multimodal AI platform combining LLMs and diffusion models, leveraged query refinement to convert raw user prompts into high-quality structured inputs, for accurate end-to-end song generation.
 - A production-grade backend architecture with async queues, serverless GPU inference, and cloud storage, ensuring efficient resource utilization and scalable performance.
- SpiroStrap** – *React.js, Express.js, Node.js, MongoDB, Redux, Recharts, Dexie, Web Bluetooth, Arduino*
 - A real-time breathing analytics system that processes high-frequency sensor data, classifies anomalies using window-based scoring, and visualizes live respiratory patterns with session playback and exports.
 - A high-performance data pipeline for continuous sensor streams, integrating local and server persistence while optimizing rendering, storage, and scalability for real-time health monitoring.

Research & Publications

- Scalable Ingestion Framework for Large Scale Time Series Data in RAG Systems:** Applied hierarchical temporal memory, correlation-aware compression, and dynamic context allocation to selectively retain high-signal patterns from time-series data, achieving 0.92 AUROC with 3× higher ingestion throughput and 56% lower latency.
- A Security Engineering Framework for H2A-PQC Integration in Heterogeneous IoT Environments:** Applied hierarchical cryptography, KEM-based authentication and Merkle tree-based signature aggregation to eliminate PQC signature bottlenecks, achieving 99.1% bandwidth reduction and 78% energy savings & enabled scalable, quantum-resilient security for large-scale heterogeneous IoT systems.

Awards and Achievements

- Won HackBattle, DEVSOC & Design Odyssey by leading teams through the complete product cycle, problem framing, MVP build, live demo, pitching and delivering stable prototypes within 48–72h hackathon windows.
- Led Assignofast's design, now serving 200+ users, delivered discovery to v1 with user research, IA/flows, and a tokenized design system.
- Earned the MongoDB Associate Database Administrator Certification, demonstrating proficiency in database management, indexing, and performance optimization.